REAL ZEROS OF $A_n(x) + kxA_{n-2}(x)$ AND A CONJECTURE OF ZHANG

KNOWLEDGE SENSE

ABSTRACT. Let $A_n(x)$ be the Eulerian polynomial of the symmetric group $\mathfrak{S}_n$. We determine, for every $n \geq 3$, the set of real numbers k for which $A_n(x) + kxA_{n-2}(x)$ has $n-1$ distinct real zeros: this happens exactly when $k < -n(n-1)$ or $k > -T_{m+1}/T_m$, where $m = \lfloor n/2 \rfloor$ and the T_m are the tangent numbers. This confirms a conjecture of Zhang. We also locate all zeros for every real k. At $k = -n(n-1)$ the number 1 is a double zero, and at $k = -T_{m+1}/T_m$ the number -1 is a multiple zero. For k strictly between these values exactly two zeros are nonreal, and they are complex conjugates of modulus one. The proof expresses A_n through polynomials F_n obtained from a constant by alternately applying two fixed first-order differential operators, and shows by elementary arguments that the zeros of F_{n-2} and F_n strictly interlace.

1. INTRODUCTION

For $n \geq 1$ let

$$A_n(x) = \sum_{\pi \in \mathfrak{S}_n} x^{\mathrm{des}(\pi)},$$

where $\mathrm{des}(\pi)$ is the number of descents of π, and put $A_0(x) = 1$. Thus $A_1 = 1$, $A_2 = 1 + x$, $A_3 = 1 + 4x + x^2$, and $\deg A_n = n - 1$ for $n \geq 1$. For $n \geq 3$ and $k \in \mathbb{R}$ we consider

$$K_{n,k}(x) = A_n(x) + kxA_{n-2}(x). \tag{1.1}$$

Polynomials of this form arise from stack sorting. Bóna [1] conjectured that the descent polynomials of t-stack sortable permutations have only real zeros, and for $t = n - 2$ the descent polynomial is $K_{n,-1}(x)$; see [9, Section 1]. Brändén [2, Theorem 5.2] proved that $K_{n,k}$ has only real and simple zeros for all $n \geq 3$ and $k > -2$; in [2] the Eulerian polynomials carry an extra factor x, which does not affect the statement. Zhang [9] gave another proof based on the **s**-Eulerian polynomials of Savage and Visontai [7], and showed that $K_{n,k}$ has only real zeros for $n > 3$ and $k \geq -n$ [9, Theorem 1.3]; the restriction $n > 3$ is needed there, since $K_{3,-3}(x) = 1 + x + x^2$. Stanley suggested determining when $K_{n,k}$ has only real zeros [9, Section 3]. Zhang first considered the case of distinct real zeros, determined the range of k numerically for $3 \leq n \leq 18$, and proposed the following conjecture.

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Let the tangent numbers T_m be defined by

$$\tan t = \sum_{m\geq 1} T_m \frac{t^{2m-1}}{(2m-1)!}, \tag{1.2}$$

so that $T_1 = 1$, $T_2 = 2$, $T_3 = 16$, $T_4 = 272$, and T_m is the number of alternating permutations of $\{1, \ldots, 2m-1\}$. This is the normalization used in [9].

Conjecture 1.1 ([9, Conjecture 3.2])**.** *For any $n \geq 3$, the polynomial $K_{n,k}(x)$ has only real and distinct zeros if and only if $k \in (-\infty, -n(n-1)) \cup (-a(\lfloor n/2 \rfloor), +\infty)$, where $a(m) = T_{m+1}/T_m$.*

Since $K_{n,k}$ has degree $n-1$ for every k, having only real and distinct zeros means having $n-1$ distinct real zeros. Our main result confirms Conjecture 1.1.

Theorem 1.2. *Let $n \geq 3$, $m = \lfloor n/2 \rfloor$ and $k \in \mathbb{R}$. Then $A_n(x) + kxA_{n-2}(x)$ has $n-1$ distinct real zeros if and only if*

$$k < -n(n-1) \qquad \text{or} \qquad k > -\frac{T_{m+1}}{T_m}.$$

Theorem 1.2 is contained in the following description of all zeros.

Theorem 1.3. *Let $n \geq 3$, $m = \lfloor n/2 \rfloor$ and $\tau_n = T_{m+1}/T_m$. Then $\tau_n < n(n-1)$. For every $k \in \mathbb{R}$ the polynomial $K_{n,k}$ has degree $n-1$ and satisfies $K_{n,k}(0) = 1$, and the following hold.*

(a) *If $k > -\tau_n$, then all zeros of $K_{n,k}$ are negative and simple.*
(b) *If $k = -\tau_n$, then -1 is a zero of multiplicity two if n is odd and of multiplicity three if n is even; all other zeros are negative and simple.*
(c) *If $-n(n-1) < k < -\tau_n$, then $K_{n,k}$ has exactly two nonreal zeros; they are simple, complex conjugate and of modulus one. All other zeros are negative and simple.*
(d) *If $k = -n(n-1)$, then 1 is a zero of multiplicity two; all other zeros are negative and simple.*
(e) *If $k < -n(n-1)$, then $K_{n,k}$ has exactly two positive zeros, both simple; all other zeros are negative and simple.*

In particular, $K_{n,k}$ has only real zeros if and only if $k \leq -n(n-1)$ or $k \geq -\tau_n$.

The last assertion of Theorem 1.3 answers the question of Stanley mentioned above. For $3 \leq n \leq 18$ the numbers $-n(n-1)$ and $-\tau_n$ are the values ω_n and Ω_n tabulated in [9, Section 3]; for instance $\tau_8 = \tau_9 = T_5/T_4 = 7936/272 = 496/17$. Zhang also observed that the products $\prod_{i=1}^{j} \Omega_{2i+1}$ are tangent numbers up to sign. This is explained by Theorem 1.3, since $\prod_{i=1}^{j} \tau_{2i+1} = \prod_{i=1}^{j} T_{i+1}/T_i = T_{j+1}$.

Method. Since A_n is palindromic, it can be written as

$$A_n(x) = (x+1)^{e_n} x^{d_n} H_n(x + x^{-1})$$

with a monic polynomial H_n in the variable $w = x + x^{-1}$ (Lemma 2.3). It is convenient to work with the variable $z = x/(x-1)^2$, which satisfies $z^{-1} = w - 2$. In this variable the recurrence of the Eulerian polynomials becomes the alternating application of the two operators

$$\mathcal{U}f = f + zf', \qquad \mathcal{T}f = (1+6z)f + z(1+4z)f'$$

to a constant (Lemma 2.1). Neither operator depends on n, so F_n and F_{n-2} are mapped to F_{n+1} and F_{n-1} by the same operator. A logarithmic-derivative argument shows that $\mathcal{U}$ and $\mathcal{T}$ preserve real-rootedness, and the Hermite–Kakeya–Obreschkoff theorem, in the special case proved in Section 3, then propagates strict interlacing from (F_{n-2}, F_n) to (F_{n-1}, F_{n+1}) (Proposition 3.4). Consequently the ratio H_n/H_{n-2} is increasing to the right of the largest zero of H_{n-2}, and only one quadratic factor of $K_{n,k}$ can leave the negative real axis. The two thresholds are the negatives of the values of this ratio at $w = 2$ and $w = -2$, that is, at $x = 1$ and $x = -1$. They are computed from $A_n(1) = n!$ and the exponential generating function of the Eulerian polynomials. Brändén's proof of [2, Theorem 5.2] likewise changes variables, by the substitution $x \mapsto x/(1+x)$, and uses a two-step recursion.

Related results. Combinations of two consecutive Eulerian polynomials have been studied from the point of view of Hurwitz stability. Yang and Zhang [8, Theorem 2.3] showed that $(x+1)A_n(x) + kxA_{n-1}(x)$ has no zeros in the open right half-plane for $n \geq 2$ and $k \geq -n$; in [8] the Eulerian polynomials are indexed differently. They also conjectured, in terms of the Euler zigzag numbers, the exact range of k for which this polynomial has no zeros in the closed right half-plane, and asked for a connection with Conjecture 1.1 [8, Conjecture 3.2 and the remark following it]. Ding and Zhu [5, Proposition 2.17] gave a criterion of this kind for polynomials satisfying a general recurrence. With $\beta = n$, $\gamma = \phi = 1$, $\nu = -1$, $\rho = 1$ and $\eta = k$ in their notation, it shows that $(x+1)A_{n+1}(x) + kxA_n(x)$ is Hurwitz stable, in their sense of having no zeros in the open right half-plane, for $n \geq 0$ and $k \geq -(n+1)$. The present paper treats the pencil (1.1), which involves A_n and A_{n-2}, and does not address the question of Yang and Zhang.

Organization. Section 2 introduces the polynomials F_n and H_n. Section 3 proves the strict interlacing of F_{n-2} and F_n. Section 4 evaluates the thresholds and proves Theorems 1.2 and 1.3.

2. The polynomials F_n and H_n

We use the classical recurrence

$$A_{n+1}(x) = (1+nx)A_n(x) + x(1-x)A_n'(x) \qquad (n \geq 0) \tag{2.1}$$

and the exponential generating function

$$\sum_{n\geq 0} A_n(x)\frac{t^n}{n!} = \frac{x-1}{x - e^{(x-1)t}}; \tag{2.2}$$

see [3]. For completeness we note that, for the series $G = \sum_{n\geq 0} A_n(x)t^n/n!$, the recurrence (2.1) is equivalent to

$$(1-xt)G_t = G + x(1-x)G_x,$$

and together with $G(x,0) = 1$ this equation determines the coefficients of G recursively. A direct computation shows that the right-hand side of (2.2) satisfies the same equation and equals 1 at $t = 0$.

For $n \geq 1$ put

$$d_n = \Big\lfloor \frac{n-1}{2} \Big\rfloor, \qquad e_n = n - 1 - 2d_n \in \{0, 1\}.$$

Thus $e_n = 0$ for odd n and $e_n = 1$ for even n. For $f \in \mathbb{R}[z]$ define

$$\mathcal{U}f = f + zf', \qquad \mathcal{T}f = (1+6z)f + z(1+4z)f',$$

and define polynomials $F_n \in \mathbb{R}[z]$ by $F_1 = 1$ and

$$F_{2m} = \mathcal{U}F_{2m-1} \quad (m \geq 1), \qquad F_{2m+1} = \mathcal{T}F_{2m} \quad (m \geq 1).$$

For example $F_2 = 1$, $F_3 = 1 + 6z$, $F_4 = 1 + 12z$ and $F_5 = 1 + 30z + 120z^2$.

Lemma 2.1. *For every $n \geq 1$,*

$$A_n(x) = (x+1)^{e_n}(x-1)^{2d_n} F_n\left(\frac{x}{(x-1)^2}\right).$$

Proof. The identity is between rational functions of x. Put $\varphi(x) = x/(x-1)^2$. Then

$$\varphi' = -\frac{x+1}{(x-1)^3}, \qquad 1 + 4\varphi = \frac{(x+1)^2}{(x-1)^2}, \qquad 1 + 6\varphi = \frac{x^2+4x+1}{(x-1)^2}. \tag{2.3}$$

We use induction on n; for $n = 1$ both sides equal 1. Assume the identity for n, and write F and F' for $F_n(\varphi)$ and $F_n'(\varphi)$.

Let $n = 2m+1$ with $m \geq 0$, so that $A_n = (x-1)^{2m}F$. By (2.3),

$$x(1-x)A_n' = -2mx(x-1)^{2m}F + x(x+1)(x-1)^{2m-2}F',$$

and (2.1) gives

$$\begin{aligned} A_{n+1} &= (x+1)(x-1)^{2m}F + x(x+1)(x-1)^{2m-2}F' \\ &= (x+1)(x-1)^{2m}\big(F + \varphi F'\big). \end{aligned}$$

This is the claim for $n+1 = 2m+2$, since $F_{2m+2} = \mathcal{U}F_{2m+1}$, $e_{2m+2} = 1$ and $d_{2m+2} = m$.

Let $n = 2m$ with $m \geq 1$, so that $A_n = (x+1)(x-1)^{2m-2}F$. By (2.3),

$$\begin{aligned} x(1-x)A_n' = &-x(x-1)^{2m-1}F - (2m-2)x(x+1)(x-1)^{2m-2}F \\ &+ x(x+1)^2(x-1)^{2m-4}F'. \end{aligned}$$

Adding $(1+2mx)A_n$, the coefficient of $(x-1)^{2m-2}F$ becomes

$$(1+2mx)(x+1) - (2m-2)x(x+1) - x(x-1) = x^2 + 4x + 1,$$

and by (2.3),

$$A_{n+1} = (x-1)^{2m}\big((1+6\varphi)F + \varphi(1+4\varphi)F'\big).$$

This is the claim for $n+1 = 2m+1$, since $F_{2m+1} = \mathcal{T}F_{2m}$, $e_{2m+1} = 0$ and $d_{2m+1} = m$. □

A real polynomial is *standard* if its leading coefficient is positive. We write I for the open interval $(-1/4, 0)$.

Lemma 2.2. *For every $n \geq 1$, F_n is standard of degree d_n, $F_n(0) = 1$, and F_n has d_n simple zeros, all in I.*

Proof. Both operators fix the constant term, so $F_n(0) = 1$. If f has degree d and leading coefficient a, then $\mathcal{U}f$ has degree d and leading coefficient $(d+1)a$, and $\mathcal{T}f$ has degree $d+1$ and leading coefficient $(4d+6)a$. This gives the degree and the sign of the leading coefficient by induction.

It remains to show that $\mathcal{U}$ and $\mathcal{T}$ map a standard polynomial f of degree d with d simple zeros in I to a polynomial whose zeros are simple and lie in I. If $d = 0$,

then $\mathcal{U}f$ is a positive constant and $\mathcal{T}f = f \cdot (1+6z)$ has the single zero $-1/6 \in I$. Let $d \geq 1$ and let $r_1 < \cdots < r_d$ be the zeros of f. Since f is standard, $f'(r_j)$ has the sign $(-1)^{d-j}$, $f(0) > 0$, and $f(-1/4)$ has the sign $(-1)^d$. Since $r_j < 0$ and $1 + 4r_j > 0$, both

$$\mathcal{U}f(r_j) = r_j f'(r_j) \qquad \text{and} \qquad \mathcal{T}f(r_j) = r_j(1+4r_j)f'(r_j)$$

have the sign $(-1)^{d-j+1}$, while $\mathcal{U}f(0) = \mathcal{T}f(0) = f(0) > 0$. Hence each of $\mathcal{U}f$ and $\mathcal{T}f$ has a zero in each of the d intervals (r_j, r_{j+1}) with $1 \leq j < d$ and $(r_d, 0)$. This accounts for all zeros of $\mathcal{U}f$. Moreover $\mathcal{T}f(-1/4) = -f(-1/4)/2$ has the sign $(-1)^{d+1}$, whereas $\mathcal{T}f(r_1)$ has the sign $(-1)^d$, so $\mathcal{T}f$ has one more zero in $(-1/4, r_1)$. This accounts for all $d+1$ zeros of $\mathcal{T}f$, and all of them lie in I. □

For $n \geq 1$ define

$$H_n(w) = (w-2)^{d_n} F_n\left(\frac{1}{w-2}\right).$$

Lemma 2.3. *For every $n \geq 1$, H_n is a monic polynomial of degree d_n,*

$$A_n(x) = (x+1)^{e_n} x^{d_n} H_n(x + x^{-1}), \tag{2.4}$$

and the zeros of H_n are the numbers $2 + \rho^{-1}$, where ρ runs over the zeros of F_n. In particular, H_n has d_n simple zeros, all less than -2.

Proof. Write $d = d_n$ and $F_n(z) = \sum_{i=0}^{d} c_i z^i$. Then $H_n(w) = \sum_{i=0}^{d} c_i (w-2)^{d-i}$, which is monic of degree d because $c_0 = F_n(0) = 1$. Since $1/\varphi(x) = x + x^{-1} - 2$, we have $(x-1)^{2d} F_n(\varphi(x)) = x^d \varphi(x)^{-d} F_n(\varphi(x)) = x^d H_n(x + x^{-1})$, and (2.4) follows from Lemma 2.1. As $H_n(2) = c_d \neq 0$, a number w is a zero of H_n if and only if $w \neq 2$ and $(w-2)^{-1}$ is a zero of F_n. The map $\rho \mapsto 2 + \rho^{-1}$ is injective and sends I into $(-\infty, -2)$, so the last statement follows from Lemma 2.2. □

3. Strict interlacing

Let $f, g \in \mathbb{R}[z]$ with $\deg f = d \geq 1$ and $\deg g = d - 1$. We say that g *strictly interlaces* f, and write $g \prec f$, if f has d simple real zeros $r_1 < \cdots < r_d$, g has $d-1$ simple real zeros $s_1 < \cdots < s_{d-1}$, and

$$r_1 < s_1 < r_2 < s_2 < \cdots < s_{d-1} < r_d.$$

For $d = 1$ the condition only says that g is a nonzero constant. The following two lemmas are special cases of the theorem of Hermite, Kakeya and Obreschkoff [6], [4], [2, Theorem 2.4]; we include proofs for completeness.

Lemma 3.1. *Let f and g be standard with $g \prec f$, $\deg f = d \geq 1$, and let $s_1 < \cdots < s_{d-1}$ be the zeros of g. Then the following hold.*

(i) *There are $c > 0$, $b \in \mathbb{R}$ and $c_1, \ldots, c_{d-1} > 0$ with*

$$\frac{f(z)}{g(z)} = cz + b - \sum_{j=1}^{d-1} \frac{c_j}{z - s_j}.$$

(ii) *The Wronskian $W = f'g - fg'$ is positive on $\mathbb{R}$.*

(iii) *For every $t \in \mathbb{R}$, the polynomial $f + tg$ has d simple real zeros. In the d open intervals into which $s_1, \ldots, s_{d-1}$ divide $\mathbb{R}$, it has exactly one zero each, and f/g is strictly increasing on each of them.*

Proof. (i) Since $\deg f = \deg g + 1$ and the zeros of g are simple, partial fractions give such an expansion with c the ratio of the leading coefficients and with residue $-c_j = f(s_j)/g'(s_j)$ at s_j. The sign of $g'(s_j)$ is $(-1)^{d-1-j}$. Exactly $d-j$ zeros of f exceed s_j, so the sign of $f(s_j)$ is $(-1)^{d-j}$. Hence $c_j > 0$.

(ii) Away from the zeros of g we have

$$W = g^2\Big(\frac{f}{g}\Big)' = g^2\Big(c + \sum_j \frac{c_j}{(z-s_j)^2}\Big) > 0,$$

and at s_j we have $W(s_j) = -f(s_j)g'(s_j) = c_j g'(s_j)^2 > 0$.

(iii) By (i), f/g is strictly increasing on each of the d intervals; on each of them it tends to $-\infty$ at the left end and to $+\infty$ at the right end. Hence $f/g = -t$ has exactly one solution in each interval. These are zeros of $f+tg$, and $f+tg$ does not vanish at s_j because $f(s_j) \neq 0$. Since $\deg(f+tg) = d$, these are all its zeros. □

Lemma 3.2. *Let p and q be standard, $\deg p = d \geq 1$ and $\deg q = d-1$. Assume that the zeros of q are real and simple, that p and q have no common zero, and that $p + tq$ has only real zeros for every $t \in \mathbb{R}$. Then $q \prec p$.*

Proof. If $d = 1$ there is nothing to prove, so let $d \geq 2$ and let $s_1 < \cdots < s_{d-1}$ be the zeros of q. Let $\mathbb{H}$ be the open upper half-plane. The function $\psi = p/q$ is holomorphic on $\mathbb{H}$. It takes no real value there: if $\psi(z) = -t$ with $t \in \mathbb{R}$ and $z \in \mathbb{H}$, then z would be a nonreal zero of $p + tq$. Since $\mathbb{H}$ is connected, $\operatorname{Im}\psi$ has constant sign on $\mathbb{H}$. As $y \to +\infty$ we have $\psi(iy) = c\,iy + O(1)$, where $c > 0$ is the ratio of the leading coefficients, so $\operatorname{Im}\psi > 0$ on $\mathbb{H}$.

Partial fractions give $\psi(z) = cz + b + \sum_j h_j/(z-s_j)$ with $b, h_j \in \mathbb{R}$, and $h_j = p(s_j)/q'(s_j) \neq 0$ because p and q have no common zero. As $\varepsilon \to 0^+$,

$$\operatorname{Im}\psi(s_j + i\varepsilon) = -\frac{h_j}{\varepsilon} + O(1),$$

so $h_j < 0$. Hence ψ has the form in Lemma 3.1(i). The argument in the proof of Lemma 3.1(iii), with $t = 0$, shows that p has exactly one zero in each of the intervals $(-\infty, s_1)$, (s_j, s_{j+1}) and (s_{d-1}, ∞). These are d simple real zeros, and $q \prec p$. □

Lemma 3.3. *If $f \in \mathbb{R}[z]$ is nonzero and has only real zeros, then $\mathcal{U}f$ and $\mathcal{T}f$ are nonzero and have only real zeros.*

Proof. Let $d = \deg f$ and let a be the leading coefficient of f. The leading coefficients of $\mathcal{U}f$ and $\mathcal{T}f$ are $(d+1)a$ and $(4d+6)a$, as in the proof of Lemma 2.2, so both are nonzero. Write $f = a\prod_j (z - r_j)$ with real r_j, repeated according to multiplicity. Let $z \notin \mathbb{R}$. Then z, $1+4z$ and $f(z)$ are nonzero, and since $(1+6z)/(z(1+4z)) = z^{-1} + \frac{1}{2}(z+\frac{1}{4})^{-1}$,

$$\frac{\mathcal{U}f(z)}{zf(z)} = \frac{1}{z} + \sum_j \frac{1}{z - r_j},$$

$$\frac{\mathcal{T}f(z)}{z(1+4z)f(z)} = \frac{1}{z} + \frac{1/2}{z+1/4} + \sum_j \frac{1}{z-r_j}.$$

For real r we have $\operatorname{Im}(z-r)^{-1} = -\operatorname{Im} z/|z-r|^2$, so the imaginary part of each right-hand side is $-\operatorname{Im} z$ times a positive number. Hence $\mathcal{U}f(z) \neq 0$ and $\mathcal{T}f(z) \neq 0$. □

Proposition 3.4. *For every $n \geq 3$ we have $F_{n-2} \prec F_n$ and $H_{n-2} \prec H_n$, and all zeros of H_n and H_{n-2} are less than -2.*

Proof. We prove $F_{n-2} \prec F_n$ by induction on n. For $n = 3$ it holds because $F_1 = 1$ and $F_3 = 1 + 6z$. Let $n \geq 3$ and assume $F_{n-2} \prec F_n$. Put $L = \mathcal{U}$ if n is odd and $L = \mathcal{T}$ if n is even. By the definition of the F_j,

$$F_{n+1} = LF_n \qquad \text{and} \qquad F_{n-1} = LF_{n-2}.$$

We write $L = u + v\,\frac{d}{dz}$; thus

$$(u, v) = (1, z) \text{ if } L = \mathcal{U}, \qquad (u, v) = \big(1 + 6z,\, z(1 + 4z)\big) \text{ if } L = \mathcal{T}.$$

Both F_n and F_{n-2} are standard by Lemma 2.2. By Lemma 3.1(iii), $F_n + tF_{n-2}$ is nonzero and has only real zeros for every $t \in \mathbb{R}$, so by Lemma 3.3 the same is true of $L(F_n + tF_{n-2}) = F_{n+1} + tF_{n-1}$. By Lemma 2.2, F_{n+1} and F_{n-1} are standard, $\deg F_{n+1} = d_{n+1} = d_{n-1} + 1 = \deg F_{n-1} + 1$, and the zeros of F_{n-1} are simple and lie in I.

Suppose that ζ is a common zero of F_{n+1} and F_{n-1}. Then $\zeta \in I$, so $v(\zeta) \neq 0$. For any f and g,

$$(Lf)g - (Lg)f = v\,(f'g - fg').$$

Applied to $f = F_n$ and $g = F_{n-2}$ at $z = \zeta$, this gives $v(\zeta)W(\zeta) = 0$, where $W = F_nF_{n-2} - F_nF'_{n-2}$. This contradicts Lemma 3.1(ii). Hence F_{n+1} and F_{n-1} have no common zero, and Lemma 3.2 gives $F_{n-1} \prec F_{n+1}$.

Now let $n \geq 3$ and $d = d_n$. By Lemma 2.3, the zeros of H_n and H_{n-2} are simple and less than -2, and they are the images of the zeros of F_n and F_{n-2} under the map $\rho \mapsto 2 + \rho^{-1}$, which is strictly decreasing on I. If $d = 1$, then $H_{n-2} = 1$ and H_n has a single simple real zero, so $H_{n-2} \prec H_n$. If $d \geq 2$, let $\rho_1 < \cdots < \rho_d$ be the zeros of F_n and $\sigma_1 < \cdots < \sigma_{d-1}$ those of F_{n-2}. Since $F_{n-2} \prec F_n$, their images satisfy

$$2 + \rho_d^{-1} < 2 + \sigma_{d-1}^{-1} < 2 + \rho_{d-1}^{-1} < \cdots < 2 + \sigma_1^{-1} < 2 + \rho_1^{-1} < -2.$$

Thus $H_{n-2} \prec H_n$. □

4. Proof of the main theorems

Lemma 4.1. *We have $H_n(2) = n!/2^{e_n}$ for $n \geq 1$. Moreover, $H_{2m+1}(-2) = T_{m+1}$ for $m \geq 0$, and $H_{2m}(-2) = T_{m+1}/2$ for $m \geq 1$.*

Proof. At $x = 1$ we have $x + x^{-1} = 2$, and $A_n(1) = n!$, so (2.4) gives the first claim.

At $x = -1$, (2.2) becomes

$$\sum_{n \geq 0} A_n(-1)\frac{t^n}{n!} = \frac{2}{1 + e^{-2t}} = 1 + \tanh t.$$

Since $\tanh t = -i \tan(it)$, (1.2) gives $\tanh t = \sum_{m \geq 0}(-1)^m T_{m+1} t^{2m+1}/(2m+1)!$. Hence $A_{2m}(-1) = 0$ for $m \geq 1$ and $A_{2m+1}(-1) = (-1)^m T_{m+1}$ for $m \geq 0$.

For $n = 2m + 1$, (2.4) reads $A_{2m+1}(x) = x^m H_{2m+1}(x + x^{-1})$, and at $x = -1$ it gives $(-1)^m T_{m+1} = (-1)^m H_{2m+1}(-2)$.

For $n = 2m$ with $m \geq 1$, (2.4) reads $A_{2m}(x) = (x + 1)x^{m-1} H_{2m}(x + x^{-1})$. Differentiating at $x = -1$, where the factor $x + 1$ vanishes, gives

$$A'_{2m}(-1) = (-1)^{m-1} H_{2m}(-2).$$

On the other hand, (2.1) at $x=-1$ with $n=2m$ gives

$$A_{2m+1}(-1) = (1-2m)A_{2m}(-1) - 2A'_{2m}(-1) = -2A'_{2m}(-1).$$

Therefore $(-1)^m T_{m+1} = 2(-1)^m H_{2m}(-2)$. □

Proof of Theorem 1.3. Fix $n \geq 3$ and write $d = d_n \geq 1$, $e = e_n$ and $m = \lfloor n/2 \rfloor$. Since $d_{n-2} = d-1$ and $e_{n-2} = e$, (2.4) gives $xA_{n-2}(x) = (x+1)^e x^d H_{n-2}(x+x^{-1})$, and therefore

$$K_{n,k}(x) = (x+1)^e x^d Q_k(x+x^{-1}), \qquad Q_k = H_n + kH_{n-2}. \tag{4.1}$$

The leading coefficient of A_n is 1 and $\deg xA_{n-2} = n-2$, so $K_{n,k}$ has degree $n-1$; also $K_{n,k}(0) = A_n(0) = 1$.

Let $R = H_n/H_{n-2}$. Since $H_{n-2} \prec H_n$ by Proposition 3.4, Lemma 3.1(iii) with $t = k$ shows that the polynomial Q_k is monic of degree d and has d simple real zeros $w_1 < \cdots < w_d$. If $d = 1$, then $H_{n-2} = 1$, $R = H_n$ is a monic linear polynomial, and we put $J = \mathbb{R}$; the only zero of Q_k is the solution w_1 of $R(w) = -k$. If $d \geq 2$, let s_{d-1} be the largest zero of H_{n-2} and put $J = (s_{d-1}, \infty)$. By Lemma 3.1(iii), the zeros $w_1, \ldots, w_{d-1}$ are less than s_{d-1}, and w_d is the unique solution of $R(w) = -k$ in J; moreover $s_{d-1} < -2$ by Proposition 3.4, so $w_1, \ldots, w_{d-1}$ are less than -2.

In both cases R is continuous and strictly increasing on J by Lemma 3.1(i), the interval J contains -2 and 2, and w_d is the unique point of J with $R(w_d) = -k$. Consequently, for $a \in \{-2, 2\}$,

$$\begin{aligned} w_d < a &\iff k > -R(a), \\ w_d = a &\iff k = -R(a), \\ w_d > a &\iff k < -R(a). \end{aligned} \tag{4.2}$$

By Lemma 4.1, and because $e_{n-2} = e_n$,

$$R(2) = \frac{n!}{(n-2)!} = n(n-1).$$

If $n = 2m+1$, then $R(-2) = H_{2m+1}(-2)/H_{2m-1}(-2) = T_{m+1}/T_m$; if $n = 2m$, then $R(-2) = (T_{m+1}/2)/(T_m/2) = T_{m+1}/T_m$. Thus $R(-2) = \tau_n$ for both parities, and $\tau_n = R(-2) < R(2) = n(n-1)$ because R is increasing on J.

Since $x^d Q_k(x+x^{-1}) = \prod_{j=1}^{d} x\,(x + x^{-1} - w_j)$, equation (4.1) becomes

$$K_{n,k}(x) = (x+1)^e \prod_{j=1}^{d} q_{w_j}(x), \qquad q_w(x) = x^2 - wx + 1. \tag{4.3}$$

The zeros of q_w have product 1 and sum w. Hence q_w has two distinct negative zeros if $w < -2$, equals $(x+1)^2$ if $w = -2$, has two complex conjugate nonreal zeros of modulus one if $-2 < w < 2$, equals $(x-1)^2$ if $w = 2$, and has two distinct positive zeros if $w > 2$. A zero x of q_w determines $w = x + x^{-1}$, so q_w and $q_{w'}$ have no common zero when $w \neq w'$, and -1 is a zero of q_w only for $w = -2$.

Since $w_1, \ldots, w_{d-1}$ are distinct and less than -2, the factors $q_{w_1}, \ldots, q_{w_{d-1}}$ contribute $2(d-1)$ distinct negative zeros, none equal to -1. If $e = 1$, the factor $x+1$ contributes the zero -1. These zeros are simple zeros of $(x+1)^e \prod_{j<d} q_{w_j}$, and none of them is a zero of q_{w_d} unless $w_d = -2$, in which case $q_{w_d} = (x+1)^2$. Therefore all zeros of $K_{n,k}$ outside those of q_{w_d} are negative and simple, and (4.2) with $R(-2) = \tau_n$ and $R(2) = n(n-1)$ gives the five cases.

If $k > -\tau_n$, then $w_d < -2$ and the zeros of q_{w_d} are negative, simple and different from -1; this is (a). If $k = -\tau_n$, then $w_d = -2$, $q_{w_d} = (x+1)^2$, and -1 has multiplicity $2 + e$; this is (b). If $-n(n-1) < k < -\tau_n$, then $-2 < w_d < 2$; this is (c). If $k = -n(n-1)$, then $w_d = 2$ and $q_{w_d} = (x-1)^2$; this is (d). If $k < -n(n-1)$, then $w_d > 2$ and q_{w_d} has two distinct positive zeros; this is (e). The final statement follows from (a) to (e). □

Proof of Theorem 1.2. By Theorem 1.3, $K_{n,k}$ has $n-1$ distinct real zeros in cases (a) and (e), and it has a multiple or a nonreal zero in cases (b), (c) and (d). □

Remark 4.2. The proof also shows how the zeros move with k. For $d \geq 2$ the numbers $w_1, \ldots, w_{d-1}$ always lie below the largest zero of H_{n-2}, so only the two zeros of q_{w_d} can leave the negative real axis. As k decreases from $+\infty$ to $-\infty$, the parameter w_d increases from s_{d-1} (from $-\infty$ if $d = 1$) to $+\infty$. The two zeros of q_{w_d} move along the negative axis until they meet at -1, where $K_{n,k}$ then has a zero of multiplicity $2 + e_n$. They then move along the unit circle until they meet at 1, where $K_{n,k}$ has a double zero, and finally they separate along the positive axis.

Use of AI-assisted tools

AI-assisted tools were used in the research and preparation of this manuscript.

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